

Front-line managers as boundary spanners: effects of span and time on nurse supervision satisfaction

RAQUEL M. MEYER RN, PhD¹, LINDA O'BRIEN-PALLAS RN, PhD, FCAHS², DIANE DORAN RN, PhD, FCAHS³, DAVID STREINER PhD⁴, MARY FERGUSON-PARÉ RN, PhD, CHE⁵ and CHRISTINE DUFFIELD RN, PhD⁶

¹Nursing Early Career Research Award Recipient, Ontario Ministry of Health and Long-Term Care and Assistant Professor (CLTA), *Lawrence S. Bloomberg Faculty of Nursing*, ²Professor, *Lawrence S. Bloomberg Faculty of Nursing and CHSRF/CIHR Chair in Nursing Health Human Resources*, ³Professor, *Lawrence S. Bloomberg Faculty of Nursing & Scientific Director, Nursing Health Services Research Unit, University of Toronto, Toronto*, ⁴Professor, *Department of Psychiatry, University of Toronto, Toronto* and ⁵Professor, *Department of Psychiatry and Behavioural Neurosciences, McMaster University, Hamilton*, ⁶Former Vice President, Professional Affairs and Chief Nurse Executive, University Health Network and Associate Professor, *Lawrence S. Bloomberg Faculty of Nursing, University of Toronto, Toronto, Canada* and ⁶Associate Dean (Research) and Director, *Centre for Health Services Management and Deputy Director, WHO Collaborating Centre for Nursing, Midwifery and Health Development, Faculty of Nursing, Midwifery and Health, University of Technology, Sydney, Australia*

Correspondence

Raquel M. Meyer
130–155 College Street
Lawrence Bloomberg Faculty
of Nursing
University of Toronto
Toronto
Ontario, Canada
M5T 1P8
E-mail: raquel.meyer@utoronto.ca

MEYER R.M., O'BRIEN-PALLAS L., DORAN D., STREINER D., FERGUSON-PARÉ M. & DUFFIELD C.
(2011) *Journal of Nursing Management* 19, 611–622

Front-line managers as boundary spanners: effects of span and time on nurse supervision satisfaction

Aim To examine the influence of nurse manager span (number of direct report staff), time in staff contact, transformational leadership practices and operational hours on nurse supervision satisfaction.

Background Increasing role complexity has intensified the boundary spanning functions of managers. Because work demands and scope vary by management position, time in staff contact rather than span may better explain managers' capacity to support staff.

Methods A descriptive, correlational design was used to collect cross-sectional survey and prospective work log and administrative data from a convenience sample of 558 nurses in 51 clinical areas and 31 front-line nurse managers from four acute care hospitals in 2007–2008. Data were analysed using hierarchical linear modelling.

Results Span, but not time in staff contact, interacted with leadership and operational hours to explain supervision satisfaction.

Conclusions With compressed operational hours, supervision satisfaction was lower with highly transformational leadership in combination with wider spans. With extended operational hours, supervision satisfaction was higher with highly transformational leadership, and this effect was more pronounced under wider spans.

Implications for Nursing Management Operational hours, which influence the manager's daily span (average number of direct report staff working per weekday), should be factored into the design of front-line management positions.

Keywords: health services research, managers, nursing leadership, satisfaction, span of control, time studies

Accepted for publication: 21 February 2011

Introduction

Nursing management roles and structures have been frequently redesigned or eliminated in response to cost-reductions, decentralization, delegation of activities to advanced practice nursing roles and a new public management paradigm (Duffield 1991, Duffield & Franks 2001, Canadian Nursing Advisory Committee 2002, Mahon & Young 2006). In many developed countries, the scope of the traditional unit-based 'head nurse' role has expanded to include operations, finances and the supervision of multidisciplinary staff across multiple areas in programme management structures, with increased role complexity reported (McGillis Hall & Donner 1997, Mahon & Young 2006, Shirey *et al.* 2008). Although integral to the retention of a satisfied and committed nursing staff (Hayes *et al.* 2006, Laschinger *et al.* 2009), the capacity of front-line managers to lead, implement and evaluate system change among staff is influenced by the scope of their role. Role expansion as a result of reorganization is compounded by an ageing managerial workforce and by recruitment challenges (Laschinger *et al.* 2008, Shirey *et al.* 2008). These trends highlight the need for evidence to effectively redesign nursing management roles.

Past research on managerial effectiveness has emphasized managers' personal attributes without necessarily considering the organizational context (Page *et al.* 2003). Span (i.e. the number of employees reporting directly to the manager) is an important structural variable that may influence staff and consumer outcomes. Wider spans have been negatively associated with new nurse retention (McCutcheon 2004), nursing work environments (McGillis Hall *et al.* 2006), worker engagement (Cathcart *et al.* 2004) and patient satisfaction (McCutcheon *et al.* 2009) in healthcare, as well as teamwork (Gittell 2001) and performance (Bohte & Meier 2001) in the airline and education sectors. Positive associations between wider spans and hospital nurse turnover (McCutcheon 2004) and staff accidents and unsafe behaviours in manufacturing settings (Hechanova-Alampay & Beehr 2001) have been documented.

However, when managerial leadership style and span are examined in combination, span has been observed to moderate the relationship between leadership and outcomes. McCutcheon *et al.* (2009) found that span moderated the relationship between managerial leadership and nurse job satisfaction and patient satisfaction; even highly transformational leaders could not overcome spans that were too wide to positively influence these outcomes. Similarly, the beneficial influence of emotionally intelligent leadership by nurse managers

on staff nurse empowerment was lessened under wider spans (Lucas *et al.* 2008). In contrast, strong leader-member exchange enhanced staff organizational commitment under wider spans in the banking industry (Schriesheim *et al.* 2000). The purpose of the present study was to examine the relationships among span, leadership and other covariates and nurse satisfaction with manager's supervision.

Theory

The framework was based on Open System Theory and the boundary spanning functions of managers. A large-scale healthcare organization is a social structure that was conceptualized as an open system comprised of supportive, maintenance, adaptive, production and management subsystems (Meyer & O'Brien-Pallas 2010). The management subsystem resolves conflicts between hierarchical levels, coordinates substructures and balances external demands with internal resources and needs (Katz & Kahn 1978). Boundary spanning, which refers to movement and coordination across internal and external system boundaries, involves negotiating the inflow and outflow of energy between subsystems and with the external environment across semi-permeable boundaries (Gittell 2003). Boundary spanning is assigned to organizational roles (e.g. managers) and may include acquiring, filtering and importing information from outside entities for distribution to internal users (Tushman & Scanlan 1981); regulating the flow of inputs and negotiating for needed resources (Ancona & Caldwell 1992); and engaging in relationship management to buffer and influence external and internal unit relations (Ancona & Caldwell 1992, Gittell 2003). Front-line managers are boundary spanners who typically coordinate and integrate inputs (e.g. information, materials and human resources), throughput processes and outputs across internal boundaries.

Supervision generates boundary spanning work for managers across subsystem, functional, spatial and hierarchical boundaries. Nurses do not provide nursing care in isolation; nursing work is reliant on the flow of inputs (e.g. patients and equipment), on other subsystems (e.g. admissions department and laboratories) and on other healthcare providers to accomplish interdependent goals. Managers coordinate across functional boundaries (e.g. professions and roles) and travel physically across spatial subsystem boundaries (e.g. departments and clinical areas) to engage in face-to-face communication, to integrate work processes and to directly supervise staff (Gittell 2003). Managers also

work across vertical boundaries in the organizational hierarchy to supervise performance (Ouchi & Dowling 1974), to ensure accountability (Jaques 1990), to create channels of appeal (Weber 1978) and to enable staff access to resources and managerial support (Blau 1968, Kanter 1977). Effective boundary spanners foster high quality communication and relationships with and between people through physical movement, interpersonal skills and conversations (Gittell 2003). Boundary spanning is therefore influenced by the number of employees reporting to the manager (i.e. span). Time in staff contact, which enables the manager to engage, supervise and coach staff (Ouchi & Dowling 1974, Gittell 2003, Kramer *et al.* 2007), also quantifies boundary spanning activity.

By understanding the vocabulary, semantics, shared beliefs and context, boundary spanners can effectively meet the contrasting information needs and demands of external areas; prevent the distortion and bias of communication across boundaries; and link internal and external areas (Tushman & Scanlan 1981). Managerial leadership, communication and support have been associated with nurse well-being, work relations and satisfaction (e.g. Greco *et al.* 2006, Hall 2007, Cummings *et al.* 2010, Hayes *et al.* 2010). Boundary spanners with work expertise better understand internal processes, demands and roles and are more often consulted by internal system users (Tushman & Scanlan 1981). Managerial experience has been associated with more fluid work styles, less overtime (Dunn & Schilder 1993) and more effective coping (Shirey *et al.* 2010) by managers.

Rationale, research objectives and study framework

Comparisons of span within and across organizations can misrepresent managerial capacity to support staff because managers may not spend the same time in staff contact because work demands and scope vary by management position (Ouchi & Dowling 1974). The purpose of the present study was to examine the influence of (1) span, as a measure of formal reporting relationships and (2) time in staff contact, as a measure of the closeness of contact between the manager and staff, on nurse satisfaction with manager's supervision. Building on previous research (Ouchi & Dowling 1974, Schriesheim *et al.* 2000, Gittell 2001, Doran *et al.* 2004), the interactions among span, time in staff contact and leadership were examined. Operational hours were investigated because of variation in work demands (Morash *et al.* 2005) and time allocation. Thus the

main effects of span, time in staff contact, leadership, operational hours and other manager covariates on nurse supervision satisfaction were examined. The interaction effects of span and time in staff contact with leadership and operational hours were tested. The extent to which models with either span or time in staff contact explained variation in nurse supervision satisfaction was determined.

The main and interaction effects examined are depicted in Figure 1. The first-order predictors reflected boundary spanner functions (i.e. span and time in staff contact) and characteristics (i.e. leadership) as well as operational hours. Manager covariates included boundary spanner characteristics (i.e. education and experience), span characteristics (i.e. administrative and clinical support roles) and control variables (i.e. manager's position tenure and worked hours). Nurse age, day shift employment, education and registration were nurse control variables because of associations with job or supervision satisfaction (Blegen & Mueller 1987, Shields & Ward 2001, Ward 2002).

Methods

Design and data collection procedures

A descriptive, correlational design was used to collect prospective survey, work log and human resource data (Table 1) between June 2007 and June 2008. The work log method was piloted. Cross-sectional survey data were collected for leadership practices, managerial job characteristics and nurse supervision satisfaction. Counts of direct reports, time in staff contact, and worked hours were collected longitudinally. Managers were surveyed about demographics (i.e. age, occupation, education, experience and tenure) and job characteristics (i.e. number of areas, support roles and

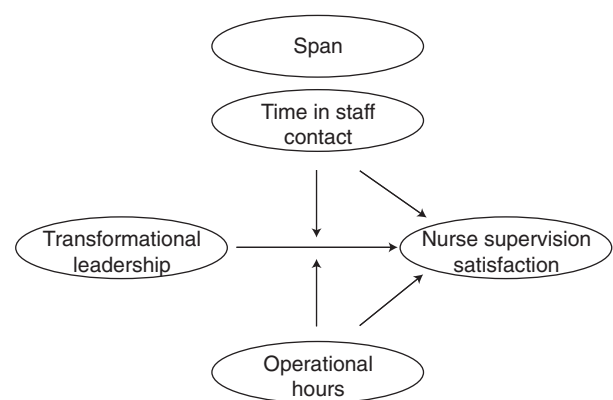


Figure 1
Framework of first-order predictors and outcome.

Table 1

Data collection overview

Sequential time line	Subject or source			
	Managers	Employees	Human resource department	Researcher
1	Pre-test of work logs			Observation of managers Inter-observer agreement
2	Survey of demographics, profession, experience, tenure, areas assigned			
3	Time study to work log staff contact Record of worked hours	Supervision satisfaction (SWMSS) Leadership practices (LPI-other) Demographics	Counts of direct reports	Observation of managers Inter-observer agreement Hours of operation, non-direct reports

LPI, Leadership Practices Inventory; SWMSS, Satisfaction with My Supervisor Scale.

budget). Subsequently, managers self-reported time in staff contact and worked hours in work logs. Surveys of nurse supervision satisfaction included demographics (e.g. age, registration, education, employment status, tenure and day shift employment). Human resource departments provided span data. Ethical approval was received from the university and participating hospitals. For managers, written consent was obtained. For nurses, a waiver of written consent was obtained and consent was assumed on receipt of the completed survey.

Sample

Acute care hospitals in a large urban city in Ontario, Canada were selected through purposive sampling. Out of six hospitals invited to participate, four agreed. Three were academic teaching hospitals and one was a large academic-affiliated community hospital. A convenience sample of managers assigned patient care units, clinics and/or services was recruited. The inclusion criteria was employment in a current front-line management position for at least 3 months with at least one direct patient care area. From across all sites, 35 out of 105 (33.3%) managers agreed to participate and of these, 31 (88.6%) completed the study. Most managers (96.8%) did not engage in direct patient care. Of a potential 1786 nurses, a convenience sample of 558 nurses (31.2%) was recruited through meetings in 51 clinical areas. The mean nurse response rate per manager was 33.6% (range: 10.8–57.7%). The inclusion criterion for nurses was employment in the current position for at least 3 months. The dataset was hierarchical with nurses grouped under managers. These two levels of analysis are hereafter identified as level-1 for nurses (at the individual level) and level-2 for managers

(at the group level). The final sample numbered 31 managers (level-2) and an average of 18 nurses (level-1) per manager. According to Scherbaum and Ferreter's (2009) estimates for hierarchical level models examining fixed effects, the sample size was adequate to achieve a power of 0.80. To reduce common method effects produced by the same participant rating the predictor (i.e. leadership) and outcome (i.e. supervision satisfaction) variables, leadership inventories for each manager (mean = 3.8, range: 2–7) were completed by a separate sub-sample ($n = 117$) of nurses and directors recruited through meetings or email.

Instrumentation

A measure of transformational leadership was selected because of its positive associations with staff autonomy (Kouzes & Posner 2002) and nurse outcomes (e.g. Cummings *et al.* 2010). Transformational leadership emphasizes how visionary, charismatic and relationship-oriented behaviours generate positive affective responses by followers and foster outstanding accomplishments (House & Aditya 1997). Transformational leadership was rated by nurses and directors familiar with the manager's behaviour using the Leadership Practices Inventory (Other Version; Posner & Kouzes 1988). Thirty items are rated on a 10-point scale (1 = almost never, 10 = almost always). Previous alpha reliabilities range from 0.81 to 0.92 (Posner & Kouzes 1993) for the five subscales (six items per subscale) which consist of: challenging the process, inspiring shared vision, enabling others to act, modelling the way and encouraging the heart. In the present study, exploratory factor analysis using principal components methods indicated a single-factor solution explaining 59.9% of the variance. A mean leadership score was

calculated to represent an over-arching construct of transformational leadership (Carless 2001). Cronbach's alpha coefficient of reliability was 0.98.

Managers completed a survey of demographics and work characteristics. Education was the highest educational qualification held by the manager (dichotomized as graduate degree *vs.* other). Experience was years worked in front-line manager positions. Position tenure was years in current position and was a proxy for the length of the relationship between the manager and staff. Managers self-reported the total number of full-time equivalent support roles in their assigned area(s) that enacted administrative (e.g. secretary and scheduling coordinator) or clinical (e.g. clinical nurse educator and discharge planner) functions.

Anonymised counts of direct reports by job title extracted at three consecutive monthly time points from human resource departments were verified by managers. Missing data (0.4%, $n = 12$) were obtained from managers. Span was calculated as the monthly average of the number of direct report employees (by headcount) assigned to the manager.

Operational hours for each assigned area were obtained from managers. A measure of weekly operational hours weighted by the number of direct reports per assigned area was created at the manager level. Operational hours were calculated per manager as the sum of the number of direct reports per area multiplied by the number of weekly operational hours per area divided by the total number of direct reports. The weighted value was then categorized as extended or compressed. Extended operational hours consisted of mainly inpatient units running 24 hours per day, 7 days a week. Compressed operational hours collapsed all managers assigned areas with purely compressed hours only (e.g. clinics) or a mix of compressed and extended areas (e.g. clinic and an inpatient unit).

The work log method and time in staff contact metrics were piloted by shadowing managers ($n = 3$). For inter-observer agreement, the first author and managers rated the presence or absence of time in staff contact at half hourly intervals ($n = 16$). Cohen's kappa coefficient, a statistical measure of inter-rater agreement, was 0.86. A time study quantified the time managers allocated to staff contact for a 1-month cycle. Managers completed work logs on 20 randomly assigned days stratified by weekday (i.e. four Mondays and four Tuesdays, etc.) over 10 weeks. On average, managers completed 18.7 work log days. Prospective, self-reported work logs were completed by managers at half hourly intervals using automatic reminders via a pager. Inter-observer agreements by the first author and each

manager evaluated the presence or absence of time in staff contact at half hourly intervals (mean = 12.7). Cohen's Kappa averaged 0.82. Time in staff contact was the average daily amount of time (in hours) spent by the manager interacting with direct and non-direct report staff, physicians and students working in the area(s) assigned to the manager. Contact included verbal, written and email communication and person-to-person interaction. Time in staff contact was calculated as the average hours in staff contact per weekday.

Worked hours were the average number of hours worked by the manager per weekday. On average, managers recorded worked hours for 34 weekdays. The count included overtime hours, but excluded on-call hours and lunch breaks. Worked hours were calculated as the sum of the worked hours per weekday divided by the number of valid weekday entries.

The Satisfaction with My Supervisor Scale (Scarpello & Vandenberg 1987) assessed nurses' satisfaction with the manager's technical, administrative and relational abilities with respect to supervision. Eighteen items were rated on a five-point Likert scale (1 = very dissatisfied, 5 = very satisfied). Cronbach's alphas in other studies range from 0.88 to 0.96 (Scarpello & Vandenberg 1987, Vandenberg & Scarpello 1992, Jones *et al.* 1999). Cronbach's alpha was 0.97 in this sample. Staff surveys included demographics for age, day shift employment (i.e. regular employment on a day shift schedule as opposed to a rotating schedule), education (i.e. highest level of nursing education achieved by the nurse) and registration [i.e. registered nurse (RN) or registered practical nurse (RPN)].

Data analyses

Statistical analyses were conducted using Statistical Package for Social Scientists (SPSS Version 16 for Windows, Chicago, IL, USA). Descriptive statistics were performed for all variables and outliers were corrected. Variables in the two-level hierarchical linear models were grand-mean centred (Raudenbush & Bryk 2002). First, a one-way analysis of variance (ANOVA) assessed within and between group variance in the outcome to determine the intra-class correlation coefficient; second, a random coefficient regression model identified significant level-1 nurse predictors; and third, intercepts-as-outcome regression models determined whether the outcome varied in relation to level-2 manager predictors for main and interaction effects among manager-level predictors. Covariates with t ratios <1 were excluded (Raudenbush & Bryk 2002). Testing of the three-way interaction involved a test of the main effects of the

predictors on the outcomes, followed by a test of the products of each pair of predictors on the outcome (i.e. two-way interactions) and then the three-way interaction (Aiken & West 1991). Span and leadership, as continuous variables, were centred on the mean prior to forming cross-product terms to minimize collinearity, and operational hours, as a categorical variable, was dummy coded (Aiken & West 1991). Because two models were tested to evaluate span and time in staff contact, Bonferroni's correction was applied.

Results

Descriptive results

Compared with study nurses, managers were older, more experienced in nursing and more likely to hold a university degree (Table 2). Compared with other Ontario managers, study managers were only slightly younger (46 *vs.* 50 years) and more likely to hold a graduate degree (39 *vs.* 8%; Canadian Institute for Health Information 2008). Study staff nurses were slightly younger (42 *vs.* 45 years) and more likely to hold a degree (35 *vs.* 27%) when compared with other Ontario nurses (Canadian Institute for Health Information 2008).

Descriptive statistics are shown in Table 3. The average span of 86.6 was slightly higher than other Ontario means of 77 (McCutcheon *et al.* 2009), 77.5 (Lucas *et al.* 2008) and 70 (Laschinger *et al.* 2008). Spans ranged from 29.0 to 174.3 direct report employees, with one-third of managers responsible for 90 or more staff. Daily time in staff contact averaged 3.2 hours (range: 1.4–7.2 hours) which was slightly higher than the 2 hours daily by Swedish healthcare managers (Arman *et al.* 2009). On average, nurse managers are engaged in leadership practices usually or

fairly often. On average, staff nurses rated their satisfaction with supervision above the midpoint of the scale range [mean = 3.82, standard deviation (SD) = 0.8] which was similar to other direct healthcare providers (Ward 2002), but somewhat higher than staff in information, government and private sectors (Keller & Dansereau 1995, Jones *et al.* 1999, Vandenberg & Nelson 1999).

Test of the study framework: main and interaction effects

Based on the variance components from the ANOVA, the intra-class correlation coefficient indicated that 18% of the variance in nurse supervision satisfaction was between managers. In the random coefficient regression model for level-1 nurse predictors, only nurse registration explained significant variation in the outcome [$t(554.87) = -2.48$, $P = 0.013$; Table 4] and was retained. Nurse age, day shift employment and education were excluded. Although RNs were less satisfied with their manager's supervision than were RPNs, grand-mean centring controlled for this effect (i.e. the outcome was adjusted for the level-1 variable; Hoffman & Gavin 1998).

In the intercepts-as-outcome models for main effects of level-2 manager first-order predictors, only leadership had a positive main effect on nurse supervision satisfaction [$t(33.14) = 3.20$, $P = 0.003$; Table 5]. Span, time in staff contact, operational hours, administrative and clinical support roles and manager experience, education, position tenure and worked hours did not exert main effects on the outcome. Overall, higher leadership scores were associated with higher supervision satisfaction scores.

In the intercepts-as-outcome models for interaction effects among level-2 manager first-order predictors, the three-way interaction with leadership and operational

	Managers (n = 31)			Nurses (n = 558)		
	%	Mean	Range	%	Mean	Range
RN	100			88		
Age		46	29–63		42	22–67
Years experience in current position/unit		3	0.3–9		8	0.25–33
Years experience in hospital		17	0.5–35		11	0.25–37
Years experience in nursing		24	6–39		16	0.33–46
Years experience in management		7	0.25–19		–	–
RPN certificate/diploma	–			9		
RN hospital school	–			2		
RN diploma	16			54		
Undergraduate degree	45			33		
Graduate degree	39			2		

RPN, registered practical nurse; RN, registered nurse.

Table 3
Descriptive statistics of study variables

<i>First-order predictors</i>	<i>%</i>	<i>Mean</i>	<i>SD</i>
Manager (Level-2)			
Span (headcount)		86.6	36.2
Time in staff contact (hours per weekday)		3.2	1.4
Transformational leadership (score range: 1–10)		7.6	1.0
Operational hours			
Compressed	38.7		
Extended	61.3		
Covariates			
Manager (Level-2)			
Education (reference: graduate degree)	38.7		
Experience (years)		6.9	5.5
Position tenure (years)		3.0	2.4
Administrative support roles* (full-time equivalents)		0.9	0.9
Clinical support roles (full-time equivalents)		3.4	1.8
Worked hours* (per weekday)		8.9	0.9
Nurse (Level-1)			
Age		42.4	10.5
Day shift employment (reference: not day shift)	73.1		
Education level (reference: college diploma)	53.9		
Registration (reference: Registered Nurse)	87.8		
Outcome			
Nurse (Level-1)			
Nurse supervision satisfaction (score range: 1–5)		3.8	0.8

*Outlier(s) identified by box plots and by z-scores >3 standard deviations from the mean (Norman & Streiner 2000) were reduced to the value of the nearest score plus one unit (Tabachnick & Fidell 1996).

Table 4
Results of fixed-coefficient regression model for main effects of level-1 nurse control variables for supervision satisfaction

<i>Parameter</i>	<i>Estimate</i>	<i>SE</i>	<i>t</i>
Estimates of fixed effects			
Intercept	3.82	0.07	53.15**
Nurse age	0.0003	0.003	0.09
Nurse day shift	0.04	0.08	0.51
Nurse education	0.03	0.07	0.47
Nurse registration	−0.28	0.11	−2.48*
Estimates of covariance parameters			
Residual	0.55	0.03	16.20***†
Intercept (subject variance)	0.13	0.04	2.99***†

* $P < 0.05$; ** $P < 0.001$; †Wald Z-test (equivalent to t -test). SE, standard error.

hours was statistically significant for span [$t(27.28) = 2.39$, $P = 0.024$; Table 6], but not for time in staff contact. The model with span explained 60.6% of the variation in nurse supervision satisfaction between managers.

The two-dimensional view of the three-way interaction for compressed *vs.* extended operational hours is presented in Figure 2 using values one standard deviation from the mean for span and leadership. When managers were assigned compressed operational hours, nurse supervision satisfaction varied by span and leadership [$t(27.28) = 2.39$, $P = 0.024$; Table 6]. As depicted in Figure 2a, under narrow spans (at 1 SD below

mean, span = 45.7), nurse supervision satisfaction was higher (4.29) with more transformational leadership (at 1 SD above mean, leadership = 8.6). Under wide spans (at 1 SD above mean, span = 134.5), nurse supervision satisfaction was lower (3.10) with more transformational leadership (at 1 SD above mean, leadership = 8.6). In other words, in spite of high leadership scores, when managers were assigned compressed operational hours in combination with wide spans, nurse supervision satisfaction scores were lower.

As illustrated in Figure 2b, when managers were assigned extended operational hours, nurses were more satisfied (3.91 *vs.* 3.67) under more transformational leadership (at 1 SD above mean, leadership = 8.6) in combination with a wider span (at 1 SD above mean, span = 115.5). The relationship between extended operational hours, leadership, span and supervision satisfaction tended to be stronger for managers with more transformational leadership [$t(29.68) = 3.00$, $P = 0.005$; Table 6]. In other words, when managers were assigned extended operational hours, transformational leadership enhanced supervision satisfaction and this effect was more pronounced under wider spans.

Discussion

The present study results partially support the boundary spanner functions and characteristics theorized to

Table 5

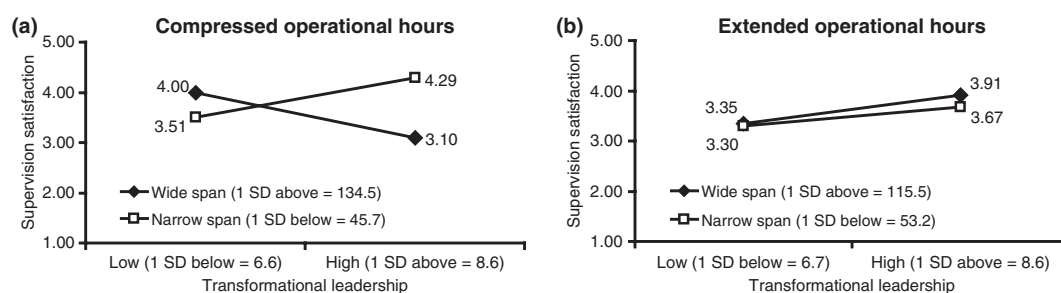
Results of intercepts-as-outcomes regression models for main effects of level-2 manager first-order variables for supervision satisfaction

Parameter	Model for span			Model for time in staff contact		
	Estimate	SE	<i>t</i>	Estimate	SE	<i>t</i>
Estimates of fixed effects						
Intercept	3.80	0.08	47.50**	3.80	0.08	46.91**
Span	-0.0008	0.002	-0.46	NA	NA	NA
Time in staff contact	NA	NA	NA	0.004	0.05	0.08
Transformational leadership	0.22	0.07	3.20**	0.22	0.07	3.14**
Operational hours	0.04	0.13	0.30	0.03	0.13	0.22
Nurse registration	-0.24	0.11	-2.22*	-0.25	0.11	-2.28*
Estimates of covariance parameters						
Residual	0.55	0.03	16.23**†	0.55	0.03	16.23**†
Intercept (subject variance)	0.09	0.03	2.82**†	0.09	0.03	2.83**†

* $P < 0.05$; ** $P < 0.01$; †Wald Z-test (equivalent to *t*-test). SE, standard error; NA, not applicable.**Table 6**

Results of intercepts-as-outcomes regression models for three-way interaction effects between span, transformational leadership and operational hours for supervision satisfaction

Parameter	Model for compressed operational hours			Model for extended operational hours		
	Estimate	SE	<i>t</i>	Estimate	SE	<i>t</i>
Estimates of fixed effects						
Intercept	3.96	0.09	42.97**	3.80	0.07	56.49**
Span	-0.004	0.002	-1.91	0.002	0.003	0.86
Transformational leadership	-0.03	0.13	-0.24	0.25	0.08	3.00**
Operational hours	-0.16	0.11	-1.43	0.16	0.11	1.43
Span × transformational leadership	-0.009	0.003	-2.79**	0.002	0.003	0.53
Span × operational hours	0.006	0.003	1.85	-0.006	0.003	-1.85
Transformational leadership × operational hours	0.28	0.15	1.84	-0.28	0.15	-1.84
Span × transformational leadership × operational hours	0.01	0.005	2.39*††	-0.01	0.005	-2.39*††
Nurse registration	-0.24	0.11	-2.22*	-0.24	0.11	-2.22*
Estimates of covariance parameters						
Residual	0.55	0.03	16.20**†	0.55	0.03	16.20**†
Intercept (subject variance)	0.05	0.02	2.24*†	0.05	0.02	2.24*†

* $P < 0.05$; ** $P < 0.01$; †Wald Z-test (equivalent to *t*-test); †† $P = 0.024$ (below Bonferroni's correction value of 0.025 for two comparisons). SE, standard error.**Figure 2**

Plots of nurse supervision satisfaction on span and transformational leadership by operational hours.

influence managerial capacity (Katz & Kahn 1978, Tushman & Scanlan 1981, Ancona & Caldwell 1992, Gittell 2003). That is, span and leadership were associated with nurse supervision satisfaction. The results

indicate that time is another type of boundary spanned by managers. The capacity of managers to influence nurse supervision satisfaction varied relative to operational hours.

As managerial work flow observations revealed, managers coordinated and integrated work processes, resources, information and relationships within and among subsystem boundaries **across time**. For example, given an average 42-hours workweek, with compressed operational hours, a manager of a 50-hours weekday clinic **covered** 84% of the operational hours and the number of staff was high during the manager's workday. That is, on any given weekday, the manager potentially interfaced with 100% of clinic staff (Figure 3a). Thus the manager's 'daily' span was nearly equivalent to the total span. In contrast, with extended operational hours, managers' worked hours did not cover most of the staff hours. For example, given an average 42-hours workweek, a manager assigned extended operational hours typically **covered** only 25% of the operational hours, yet was responsible for work processes and staff around-the-clock (Figure 3b). Extended operational hours lower the number of direct report staff working during the manager's workday. That is, on any given weekday, the manager only interfaced with approximately 7% (1/14) of unit staff (assuming 12-hours shifts). Thus the manager's 'daily' span was much lower than the total span. The study findings highlight that time, in the form of operational hours, is another boundary that managers must manage to ensure organizational functioning. Because operational hours influence not only the hours covered by the manager, but also the daily span of the manager, operational hours are an important factor to consider for front-line management positions. This finding extends the boundary spanning literature with regard to temporal boundaries.

The key finding for nurse supervision satisfaction was that the relationship between span and leadership varied by operational hours. When managers had highly transformational leadership styles and were assigned compressed operational hours in combination with wide spans, nurse supervision satisfaction was lower. This interaction makes sense because with compressed operational hours, the manager's daily span is high. One

interpretation is that with compressed operational hours, large numbers of staff may overwhelm the capacity of even highly transformational managers to positively influence supervision satisfaction. No comparative studies examining these relationships were located.

Alternatively, this interaction may indicate that some nurses were more satisfied with supervision when managers with compressed operational hours had less transformational leadership styles in combination with wide spans. Under wide spans, managers could have engaged in other types of behaviours (e.g. more task-focused leadership) to resolve work process and relational issues that impeded nurses' workflow, hence enhancing nurse supervision satisfaction. In the present study, managers assigned compressed operational hours often oversaw areas characterized by high volume throughput (e.g. clinic visits and procedures). Shorter and frequent work cycles intensify coordination demands between work roles and groups. Nurses may have been more satisfied with supervision when managers focused more on tasks or managed by exception, which could have, in turn, enhanced nurses' workflow and productivity. The leadership instrument used in the present study was developed based on project leaders. A possible conjecture is that other leadership styles may be effective when managing high volume operations. Although person-focused leadership, including transformational leadership, has been frequently associated with higher staff satisfaction, equivocal or positive results have also been observed in relation to task-focused leadership (Cummings *et al.* 2010). In the present study, task-focused behaviours may have been conducive to meeting nurses' needs for supervision when managers with compressed operational hours managed wide spans in areas with high volume throughput.

In contrast, with extended operational hours, nurses were more satisfied with supervision when managers were highly transformational, and this effect was even more pronounced under wider spans. Although

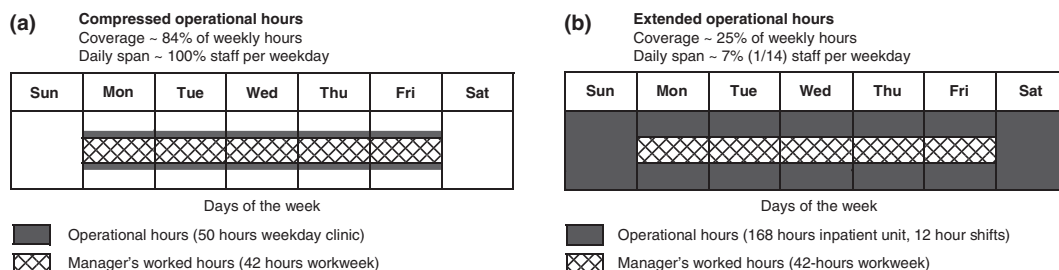


Figure 3
Manager hours relative to weekly operational hours.

counterintuitive, this interaction illustrates the importance of operational hours. With extended operational hours, narrow spans resulted in very low daily spans, whereas wide spans produced more moderate daily spans. In the present study, extended operational hours combined with wide spans may have enabled highly transformational managers to interface with more staff during weekdays (i.e. more moderate daily spans) to resolve issues. This is consistent with Blau (1968) who theorized that access to managerial support for resolution of complex work problems is a central underlying mechanism of an effective reporting structure between managers and skilled professionals. The present study result is similar to Schriesheim *et al.*'s (2000) observation that stronger supervisory leadership was associated with higher staff organizational commitment under wider spans.

However, the results are dissimilar to two studies which reported the negative moderating effect of wide spans on nurse job satisfaction (McCutcheon *et al.* 2009) and empowerment (Lucas *et al.* 2008) in spite of strong person-focused leadership. A possible explanation for the contrary finding is that the outcomes of nurse job satisfaction and empowerment are broader concepts that reflect multi-faceted aspects of nurses' work experiences. For example, workload, work-life balance and opportunities for professional development may not necessarily be amenable to intervention by the manager, but may be more sensitive to other factors (e.g. availability of organizational resources). In comparison to job satisfaction and empowerment, nurse supervision satisfaction is a narrower concept focused on the direct-report relationship.

The present study results suggest that trade-offs may exist in the staff outcomes associated with wide *vs.* narrow spans. That is, outcomes such as supervisor or organizational commitment and organizational citizenship behaviours could be theorized to improve under wide spans in combination with extended operational hours. Replication of the study design is warranted. Future studies should consider the temporal boundaries (i.e. operational hours) assigned to managers as well as outcomes sensitive to managerial coverage and to daily spans.

Although managers require time to interact, to build relationships and to coach staff (Ouchi & Dowling 1974, Gittell 2003, Kramer *et al.* 2007), time in staff contact did not predict nurse satisfaction with manager's supervision. However, time in staff contact may interact with span and this was not tested; a four-way interaction that includes span, or a sample with equivalent operational hours, may clarify these relationships.

Alternatively, Arman *et al.* (2009) noted that managers may spend time with the same, instead of varied, employees. Future studies of time in staff contact may benefit from more detailed measurement of social relationships (e.g. social network analysis).

Limitations

The cross-sectional study design limits claims between cause and effect. Although managers were less tenured, less experienced and more likely to hold graduate degrees than their counterparts in previous Canadian studies (Laschinger *et al.* 2008, Lucas *et al.* 2008, McCutcheon *et al.* 2009), these covariates did not explain variation in nurse supervision satisfaction. The potential response bias of self-report measures from a convenience sample must be considered. Common method bias was reduced by collecting data using different raters, sources and media.

Although statistically significant, the three-way interaction ought to be interpreted cautiously and warrants further investigation. The present finding may have occurred by chance; however, Bonferroni's correction was applied and the power to detect differences is lessened with higher order terms in interactions (Aiken & West 1991). The analysis accounted for clustering effects of staff under managers. Unit-level variation was not considered because one-third of managers had only one unit, and too few hospitals participated to consider the influence of hospital-level variables.

Implications for nursing management

In the present study, transformational leadership positively influenced nurse supervision satisfaction. Given that transformational leadership practices can be learned (Kouzes & Posner 2002), managers and organizations are encouraged to continue to invest in professional development opportunities to enhance leadership skills. This strategy maximizes existing managerial resources without increasing the number of managers. However, neither leadership, nor span, can be considered in isolation. Operational hours also need to be factored into front-line management job descriptions. In the present study, when managers were highly transformational leaders, moderate daily spans for nurse supervision satisfaction were enabled by narrow spans with compressed operational hours and by wide spans with extended operational hours.

Co-manager models, which have been recommended to improve manager outcomes and staff retention

(Carroll *et al.* 2004, Shirey *et al.* 2008, 2010), could be considered when compressed operational hours are combined with very wide spans. This strategy creates slack in the system by importing additional managerial resources. The trade-offs of co-manager models are likely to include dual reporting relationships, new coordination demands between co-managers (e.g. communication and consistency) and added managerial salary costs. However, additional cost savings related to supervision satisfaction and other process improvements may arise.

Alternatively, to optimize managerial coverage and daily span, managers who bridge shifts during their workday or who also work evening, night and weekend shifts are likely to cover more operational hours and interface with more staff. These strategies maximize existing managerial resources. However, co-manager models would likely better support shift rotations by managers.

Conclusion

As healthcare organizations contend with staffing shortages, front-line managers are critical to the retention and supervision of human resources. The span and operational hours assigned to managers affect their capacity to influence nurse supervision satisfaction. Although outcomes may improve under narrower (or wider) spans, the interactive effects of span, leadership, operational hours and other covariates merit consideration. Additional research is needed to explicate the conditions under which managers influence organizational, staff and healthcare consumer outcomes.

Acknowledgements

This work was supported by doctoral fellowships from the Canadian Institutes of Health Research (No. 70487) and the Nursing Health Services Research Unit (University of Toronto).

References

- Aiken L.S. & West S.G. (1991) *Multiple Regression: Testing and Interpreting Interactions*. Sage Publications, Newbury Park, CA.
- Ancona D.G. & Caldwell D.F. (1992) Bridging the boundary: external activity and performance in organizational teams. *Administrative Science Quarterly* 37, 634–665.
- Arman R., Dellve L., Wikström E. & Törnström L. (2009) What health care managers do: applying Mintzberg's structured observation method. *Journal of Nursing Management* 17, 718–729.
- Blau P.M. (1968) The hierarchy of authority in organizations. *American Journal of Sociology* 73, 453–467.
- Blegen M.A. & Mueller C.W. (1987) Nurses' job satisfaction: a longitudinal analysis. *Research in Nursing and Health* 10, 227–237.
- Bohte J. & Meier K.J. (2001) Structure and the performance of public organizations: task difficulty and span of control. *Public Organization Review* 1, 341–354.
- Canadian Institute for Health Information (2008) Regulated nursing workforce by position and province/territory of registration, 2007. [WWW document]. URL Available at: http://secure.cihi.ca/cihiweb/disPage.jsp?cw_page=statistics_nursing_2007_e#k.
- Canadian Nursing Advisory Committee (2002) *Our Health, Our Future: Creating Quality Workplaces for Canadian Nurses*. Advisory Committee on Health Human Resources Ottawa, Ontario, Canada.
- Carless S.A. (2001) Assessing the discriminant validity of the Leadership Practices Inventory. *Journal of Occupational and Organizational Psychology* 74, 233–239.
- Carroll C.A., Lacey S.R. & Cox K.S. (2004) Comparing variations in labor costs for two vs. one full-time nurse manager. *Nursing Economics* 22, 254–257.
- Cathcart D., Jeska S., Karnas J., Miller S.E., Pechacek J. & Rheault L. (2004) Span of control matters. *Journal of Nursing Administration* 34, 395–399.
- Cummings G.G., MacGregor T., Davey M. *et al.* (2010) Leadership styles and outcome patterns for the nursing workforce and work environment: a systematic review. *International Journal of Nursing Studies* 47, 363–385.
- Doran D., McCutcheon A.S., Evans M.G. *et al.* (2004) *Impact of the Manager's Span of Control on Leadership and Performance*. Canadian Health Services Research Foundation, Ottawa, ON.
- Duffield C. (1991) First-line nurse managers: issues in the literature. *Journal of Advanced Nursing* 16, 1247–1253.
- Duffield C. & Franks H. (2001) The role and preparation of first-line nurse managers in Australia: where are we going and how do we get there? *Journal of Nursing Management* 9, 87–91.
- Dunn K. & Schilder E. (1993) Novice and expert head nurses: a comparative study of work activities and behaviours. *Canadian Journal of Nursing Administration* 6, 10–15.
- Gittell J.H. (2001) Supervisory span, relational coordination and flight departure performance: a reassessment of postbureaucracy theory. *Organization Science* 12, 468–483.
- Gittell J.H. (2003) A theory of relational coordination. In *Positive Organizational Scholarship: Foundations of a New Discipline* (K.S. Cameron, J.E. Dutton & R.E. Quinn eds), pp. 279–435. Berrett-Koehler Publishing, San Francisco, CA.
- Greco P., Laschinger H.K.S. & Wong C. (2006) Leader empowering behaviours, staff nurse empowerment and work engagement/burnout. *Canadian Journal of Nursing Leadership* 19, 41–56.
- Hall D.S. (2007) The relationship between supervisor support and Registered Nurse outcomes in nursing care units. *Nursing Administration Quarterly* 31, 68–80.
- Hayes L.J., O'Brien-Pallas L., Duffield C. *et al.* (2006) Nurse turnover: a literature review. *International Journal of Nursing Studies* 43, 237–263.
- Hayes B., Bonner A. & Pryor J. (2010) Factors contributing to nurse job satisfaction in the acute hospital setting: a review of recent literature. *Journal of Nursing Management* 18, 804–814.

- Hechanova-Alampay R. & Beehr T.A. (2001) Empowerment, span of control, and safety performance in work teams after workforce reduction. *Journal of Occupational Health Psychology* 6, 275–282.
- Hoffman D.A. & Gavin M.B. (1998) Centering decisions in hierarchical linear models: implications for research in organizations. *Journal of Management* 24, 623–641.
- House R.J. & Aditya R.N. (1997) The social scientific study of leadership: quo vadis? *Journal of Management* 23, 409–473.
- Jaques E. (1990) In praise of hierarchy. *Harvard Business Review* 68, 127–133.
- Jones F.F., Scarpello V. & Bergmann T. (1999) Pay procedures – what makes them fair? *Journal of Occupational and Organizational Psychology* 72, 129–145.
- Kanter R.M. (1977) *Men and Women of the Corporation*. Basic Books, New York, NY.
- Katz D. & Kahn R.L. (1978) *The Social Psychology of Organizations*. John Wiley & Sons, New York, NY.
- Keller T. & Dansereau F. (1995) Leadership and empowerment: a social exchange perspective. *Human Relations* 48, 127–146.
- Kouzes J.M. & Posner B.Z. (2002) *The Leadership Challenge*, 3rd edn. Jossey-Bass, San Francisco, CA.
- Kramer M., Maguire P., Schmalenberg C. *et al.* (2007) Nurse manager support: what is it? Structures and practices that promote it. *Nursing Administration Quarterly* 31, 325–340.
- Laschinger H.K.S., Wong C., Ritchie J. *et al.* (2008) A profile of the structure and impact of nursing management in Canadian hospitals. *Healthcare Quarterly* 11, 85–94.
- Laschinger H.K.S., Finegan J. & Wilk P. (2009) Context matters: the impact of unit leadership and empowerment on nurses' organizational commitment. *Journal of Nursing Administration* 39, 228–235.
- Lucas V., Laschinger H.K.S. & Wong C. (2008) The impact of emotional intelligent leadership on staff nurse empowerment: the moderating effect of span of control. *Journal of Nursing Management* 16, 964–973.
- Mahon A. & Young R. (2006) Health care managers as a critical component of the health care workforce. In *Human Resources for Health in Europe* (C. Dubois, M. McKee & E. Nolte eds), pp. 116–139. Open University Press, Berkshire.
- McCutcheon A.S. (2004) *Relationships between leadership style, span of control and outcomes*. Unpublished doctoral thesis, Graduate Department of Nursing Science, University of Toronto, Toronto, ON.
- McCutcheon A.S., Doran D., Evans M., McGillis Hall L. & Pringle D. (2009) Effects of leadership and span of control on nurses' job satisfaction and patient satisfaction. *Canadian Journal of Nursing Leadership* 22, 48–67.
- McGillis Hall L. & Donner G.J. (1997) The changing role of hospital nurse managers: a literature review. *Canadian Journal of Nursing Administration* 10, 14–39.
- McGillis Hall L., Doran D., O'Brien-Pallas L. *et al.* (2006) *Quality Worklife Indicators for Nursing Practice Environments in Ontario: Determining the Feasibility of Collecting Indicator Data*. University of Toronto, Toronto, ON.
- Meyer R.M. & O'Brien-Pallas L.L. (2010) Nursing Services Delivery Theory: an open system approach. *Journal of Advanced Nursing* 66, 2828–2838.
- Morash R., Brintnell J. & Rodger G.L. (2005) A span of control tool for clinical managers. *Canadian Journal of Nursing Leadership* 18, 83–93.
- Norman G.R. & Streiner D.L. (2000) *Biostatistics: The Bare Essentials*, 2nd edn. B.C. Decker Inc., Hamilton, Ontario, Canada.
- Ouchi W.G. & Dowling J.B. (1974) Defining the span of control. *Administrative Science Quarterly* 19, 357–365.
- Page C., Wilson M., Meyer D. & Inkson K. (2003) "It's the situation I'm in": the importance of managerial context to effectiveness. *Journal of Management Development* 22, 841–862.
- Posner B.Z. & Kouzes J.M. (1988) Development and validation of the Leadership Practices Inventory. *Educational and Psychological Measurement* 48, 483–496.
- Posner B.Z. & Kouzes J.M. (1993) Psychometric properties of the Leadership Practices Inventory – updated. *Educational and Psychological Measurement* 53, 191–199.
- Raudenbush S.W. & Bryk A.S. (2002) *Hierarchical Linear Models: Applications and Data Analysis Methods*. Sage Publications, Thousand Oaks, CA.
- Scarpello V. & Vandenberg R.J. (1987) The satisfaction with my supervisor scale: its utility for research and practical applications. *Journal of Management* 13, 447–466.
- Scherbaum C.A. & Ferreter J.M. (2009) Estimating statistical power and required sample sizes for organizational research using multilevel modeling. *Organizational Research Methods* 12, 347–367.
- Schriesheim C.A., Castro S.L. & Yammarino F.J. (2000) Investigating contingencies: an examination of the impact of span of supervision and upward controllability on leader-member exchange using traditional and multivariate within- and between-entities analysis. *Journal of Applied Psychology* 85, 659–677.
- Shields M.A. & Ward M. (2001) Improving nurse retention in the National Health Service in England: the impact of job satisfaction on intentions to quit. *Journal of Health Economics* 20, 677–701.
- Shirey M.R., Ebright P.R. & McDaniel A.M. (2008) Sleepless in America: nurse managers cope with stress and complexity. *Journal of Nursing Administration* 38, 125–131.
- Shirey M.R., McDaniel A.M., Ebright P.R., Fisher M.L. & Doebbeling B.N. (2010) Understanding nurse manager stress and work complexity: factors that make a difference. *Journal of Nursing Administration* 40, 82–91.
- Tabachnick B.G. & Fidell L.S. (1996) *Using Multivariate Statistics*, 3rd edn. Harper Collins College Publishers, New York, NY.
- Tushman M.L. & Scanlan T.J. (1981) Boundary spanning individuals: their role in information transfer and their antecedents. *Academy of Management Journal* 24, 289–305.
- Vandenberg R.J. & Nelson J.B. (1999) Disaggregating the motives underlying turnover intentions: when do intentions predict behavior? *Human Relations* 52, 1313–1336.
- Vandenberg R.J. & Scarpello V. (1992) Multitrait-multimethod validation of the satisfaction with my supervisor scale. *Educational and Psychological Measurement* 52, 203–212.
- Ward L.W. (2002) *A Survey and Description of Direct Care Providers' Satisfaction with their Supervisors: A Look at Race, Education, Gender, and Age Characteristics of Direct Care Providers*. Unpublished doctoral thesis. Sargent College of Health and Rehabilitation Sciences, Boston University, Boston, MA.
- Weber M. (1978) *Economy and Society: An Outline of Interpretive Sociology* (E. Fischhoff, H. Gerth, A.M. Henderson *et al.* Trans. Vol. 2). University of California Press, Berkeley, CA.